32-bit Modular Arithmetic and SIMD Vectorisation

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1 Introduction

This project explores and optimize modular reductions for prime numbers $p < 2^{32}$ using SIMD Vectorisation (AVX2, AVX512, NEON) along with various different techniques and algorithmes.

Objective: The aim of this study is to be able to realize an optimized implementation with minimal performance cost for modular reductions.

2 Fundamental Cost of Modular Reduction

The most naive way to compute $a \pmod{n}$ is to perform the operation :

$$a - \left\lfloor \frac{a}{n} \right\rfloor \cdot n \tag{1}$$

However, on modern processor architectures (Intel, AMD,...), integer division has significantly higher computational cost than other operations like additions and multiplications.

Since equation (1) requires a divsion and substraction operation, its considered highly inefficient from both a software and hardware perspective.

3 Shoup's Modular Multiplication

Shoup's algorithm is centraled around precomputation as a work around the costly division operations

Algorithm 1: Shoup's modular multiplication algorithm

Input : $A, B \in [0, p[$, we suppose that $p < \beta/2$
precompute $B' = \lfloor B\beta/p \rfloor$

Output: $R = AB \bmod p$

$Q \leftarrow \lfloor AB'/\beta \rfloor$;
$R \leftarrow (AB - Qp) \bmod \beta$;
if $R \geq p$ **then**
  $R \leftarrow R - p$;

3.1 Input Constraint:

To ensure the correctness of the modular reduction and prevent register overflow during SIMD operations, the following conditions must be satisfied:

- **Word Size (β):** Set to 2^{32} for 32-bit modular arithmetic
- **Input 1 (A):** Integer defined over 32 bits
- **Input 2 (B):** Integers within the range $[0, p[$
- **Modulus (p):** A prime number defined over 31 bits ($p < 2^{31}$)

3.2 Theorem

Let p be a prime number such that $p < \beta/2$ where $\beta = 2^{32}$. Given integers A an integer defined over 32 bits and $B \in [0, p)$, let B' be the precomputed value $B' = \lfloor B\beta/p \rfloor$. If the estimated quotient is defined as $Q = \lfloor AB'/\beta \rfloor$, then the intermediate remainder $R = AB - Qp$ satisfies the following condition:

$$0 \leq AB - Qp < 2p$$

3.3 Input Hypothesis:

Can p be any integer over 32 bits?
No, one particularly interesting case where Shoup's algorithme will fail is for $p \geq 2^{31}$. Since we know that:

$$0 \leq AB - Qp < 2p$$

if $p \geq 2^{31}$ thus $2p \geq 2^{32}$. Therefore, in some cases $AB - Qp$ may exceed 2^{32}. This causes tremendous errors in the calculation because all bits beyond 32 bits are destroyed when we mod β

Does p have to neccesarily be a prime number?
No, the algorithm mainly uses precomputations (Barrett Reduction), optimized for caculation of $a \mod n$ without constraints on n. But for the sake of this project, n is a prime number (p).

Can b be any integer over 32 bits?
No, suppose $b > p$, we can rewrite b as:

$$b = q' \cdot p + r'$$

Thus the caculations for the precomputed value B' is:

$$B' = \left\lfloor \frac{(q' \cdot p + f) \cdot 2^{32}}{p} \right\rfloor$$

$$B' = q' \cdot 2^{32} + \left\lfloor \frac{f \cdot 2^{32}}{p} \right\rfloor$$

With a focus on 32 bits in this project, B' in the case of $b > p$ cannot be stored within 32 bits and will overflow.
Can a be any integer over 32 bits?
Yes, a can be any integer defined over 32 bits

3.4 Procomputational Logic (Barrett Reduction)

The precomputed value B' is defined over 32 bits as:

$$B' = \lfloor (B \cdot 2^{32})/p \rfloor$$

When computing the $a \cdot b \pmod{p}$:

$$A \cdot B \pmod{p} = A \cdot B - \left\lfloor \frac{A \cdot B}{p} \right\rfloor \cdot p$$

We can rewrite this:

$$\begin{aligned} A \cdot B \pmod{p} &= A \cdot B - \left\lfloor \frac{A \cdot \left\lfloor (\frac{B \cdot 2^{32}}{p}) \right\rfloor}{2^{32}} \right\rfloor \cdot p \\ &\approx A \cdot B - \left\lfloor \frac{A \cdot B'}{2^{32}} \right\rfloor \cdot p \end{aligned}$$

3.5 Fundamental Principal

Given x the quotient of 2 32 bits integers:
The proof of this algorithm based on the non decimal value of ($\lfloor x \rfloor$):

$$x - 1 < \lfloor x \rfloor \leq x$$

We can rewrite as:

$$0 \leq x - \lfloor x \rfloor < 1$$

3.6 Demonstration:

The algorithm uses the precomputation $B' = \lfloor B\beta/p \rfloor$ such that:

$$0 \leq \frac{B\beta}{p} - \left\lfloor \frac{B\beta}{p} \right\rfloor < 1$$

$$0 \leq \frac{B\beta}{p} - B' < 1 \tag{2}$$

Using the same logic, the algorithm estimates the quotient $Q = \lfloor AB'/\beta \rfloor$ such that:

$$0 \leq \frac{AB'}{\beta} - \left\lfloor \frac{AB'}{\beta} \right\rfloor < 1$$

$$0 \leq \frac{AB'}{\beta} - Q < 1 \tag{3}$$

We now prouve that remainder $R = (A \cdot B - Q \cdot p)$ is within the interval [0, 2p[:

We firstly multiply the equation (1) by Ap/β:

$$0 \leq \left(\frac{B\beta}{p} - B'\right) \cdot \frac{Ap}{\beta} < 1 \cdot \frac{Ap}{\beta}$$

$$0 \leq AB - \frac{B'Tp}{\beta} < \frac{Ap}{\beta} \tag{4}$$

Similarly, we multiply the equation (2) by p:

$$0 \leq \left(\frac{AB'}{\beta} - Q\right) \cdot p < 1 \cdot p$$

$$0 \leq \frac{AB'p}{\beta} - Qp < p \tag{5}$$

By adding the equation (3) and (4), we deduce:

$$0 \leq (AB - \cancel{\frac{AB'p}{\beta}}) + (\cancel{\frac{AB'p}{\beta}} - Qp) < \frac{Ap}{\beta} + p$$

Simplifying this result, we are left with:

$$0 \leq AB - Qp < p + \frac{Ap}{\beta}$$

$$0 \leq AB - Qp < p \cdot (1 + \frac{A}{\beta})$$

By constraint, $A \in [0, p[$ and $p < \beta$ thus $A < \beta$. Therefore we can rewrite:

$$0 \leq AB - Qp < p \cdot (1 + \frac{A}{\beta}) < p \cdot (1 + \frac{\beta}{\beta})$$

$$0 \leq AB - Qp < p \cdot (1 + 1)$$

$$0 \leq AB - Qp < 2p$$

$$0 \leq R < 2p$$

Conclusion: We have now prouven that $R \in [0, 2p[$ the final if statement of the helps correct the remainder. If $R > p$ the modular reduction needs 1 more step $R - p$, if not R is the correct result

3.7 Avantages

Using the precomputed value B' to approxiamtely calculate $\left\lfloor \frac{A \cdot B}{p} \right\rfloor$ improves on the naive method by 2 ways:

1. **Storing the Inverse of B:** B' is computed with 1 division and can be stored to be used in other modular reduction provided the same B provided the same p
2. **Shoup's algorithm replaces division operations (slow) with multiplications and additions (fast):** $\frac{A \cdot B'}{2^{32}}$ has only 1 multiplication operation and a bit shift to the right of 32 bits. On a CPU, both multiplication and bit shift operations can be done efficiently compared to division

3.8 SIMD Vectorization Implementation

We decided to create 2 versions for implementating a high performance Shoup algorithm

3.8.1 AVX

For the AVX version, we decided to use 2 main AVX function: _mm256_mul_epu32 and _mm256_mul_epi32.

Intel _mm256_mul_epu32

Listing 1: Auxilary Function of Shoup's Modular Reduction

```
    static inline __m256i _shoup_avx(__m256i va,
                                     __m256i vb,
                                     __m256i vb_precomp,
                                     __m256i vp,
                                     __m256i mask32)
{
    // 1. q = (a * b_precomp) >> 32:
    __m256i q = _mm256_mul_epu32(va, vb_precomp);
    q = _mm256_srli_epi64(q, 32);

    // 2. ab:
    __m256i ab = _mm256_mul_epu32(va, vb);

    // 3. qp:
    __m256i qp = _mm256_mul_epu32(q, vp);

    // 4. c = (ab - qp) & mask:
    __m256i c = _mm256_sub_epi64(ab, qp);
    c = _mm256_and_si256(c, mask32);

    // 5. if (c >= p) c -= p
    __m256i compare = _mm256_cmpgt_epi64(vp, c);
    __m256i sub_mask = _mm256_andnot_si256(compare, vp);

    return _mm256_sub_epi64(c, sub_mask);
}
```

Listing 2: AVX2 Implementation of Shoup's Modular Reduction

```
    static Vector _shoup_scale_avx(Parameters param, Vector v)
{
    /*1. Preloading b, b_precomp, p and 32 bit mask*/
    int size = v.size;
    Vector res = init_vector(size);
    __m256i vb = _mm256_set1_epi64x(param.b);
    __m256i vb_precomp = _mm256_set1_epi64x(param.b_precomp);
    __m256i vp = _mm256_set1_epi64x(param.p);
    __m256i mask_32 = _mm256_set1_epi64x(0xFFFFFFFF);

    int i = 0;
    for (; i + 7 < size; i += 8)
    {
        /* 2. Load 8 elements [v7 v6 v5 v4 v3 v2 v1 v0]: */
        __m256i va_input = _mm256_loadu_si256((__m256i const *)
                                (v.elements));
```

```
        /* 3. Split into two sets of 4 : va_input and va_odd */
        // Elements [null, 7, null, 5, null , 3, null, 1]
        __m256i va_odd = _mm256_srli_epi64(va_input, 32);

        /* 4. Process Even Lanes: */
        __m256i c_even = _shoup_avx(va_input, vb,
                                    vb_precomp, vp, mask_32);

        /* 5. Process Odd Lanes: */
        __m256i c_odd = _shoup_avx(va_odd, vb, vb_precomp,
                                  vp, mask_32);

        /* 6. Shift odd by 32 bits to the left results back
        and merge: */
        // [7, null, 5, null, 3, null, 1, null]
        __m256i res_odd_shifted = _mm256_slli_epi64(c_odd, 32);

        // [7, null, 5, null, 3, null, 1, null]
        //or null, 6, null, 4, null, 2, null, 0]
        __m256i res_combined = _mm256_or_si256(c_even,
                                               res_odd_shifted);
        /* 7. Store: */
        _mm256_storeu_si256((__m256i *)(res.elements),
                            res_combined);
    }
    for (; i < size; i++)
        res.elements[i] = shoup(v.elements[i], param.b,
                                param.b_precomp, param.p);
    return res;
}
```

Intel _mm256_mullo_epi32

Listing 3: Auxilary Function of Shoup's Modular Reduction (Version mullo)

```
    static inline __m256i _shoup_avx_mullo(__m256i va,
                                           __m256i vb,
                                           __m256i vb_precomp,
                                           __m256i vp)
{
    // 1. q = (a * b_precomp) >> 32:
    __m256i q = _mm256_mul_epu32(va, vb_precomp);
    q = _mm256_srli_epi64(q, 32);

    // 2. ab:
    __m256i ab = _mm256_mullo_epi32(va, vb);

    // 3. qp
    __m256i qp = _mm256_mullo_epi32(q, vp);

    // 4. c = ab - qp
    __m256i c = _mm256_sub_epi32(ab, qp);

    // 5. if (c >= p) c -= p
    __m256i compare = _mm256_cmpgt_epi64(vp, c);
    __m256i sub_mask = _mm256_andnot_si256(compare, vp);

    return _mm256_sub_epi64(c, sub_mask);
}
```

Listing 4: VX2 Implementation of Shoup's Modular Reduction (Version mullo)

```
    static Vector _shoup_scale_avx_mullo(Parameters param, Vector v)
{
    /*1. Preloading b, b_precomp, p and 32 bit mask*/
    int size = v.size;
    Vector res = init_vector(size);
    __m256i vb = _mm256_set1_epi64x(param.b);
    __m256i vb_precomp = _mm256_set1_epi64x(param.b_precomp);
    __m256i vp = _mm256_set1_epi64x(param.p);

    int i = 0;
    for (; i + 31 < size; i += 32)
    {
        /* 2. Load 8 elements [v7 v6 v5 v4 v3 v2 v1 v0]: */
        __m256i va_input_1 = _mm256_loadu_si256((__m256i const *)
                                                 (v.elements));

        /* 3. Split into two sets of 4 : va_input and va_odd */
        // Elements [null, 7, null, 5, null , 3, null, 1]
        __m256i va_odd = _mm256_srli_epi64(va_input, 32);

        /* 4. Process Even Lanes: */
        __m256i c_even = _shoup_avx_mullo(va_input, vb,
                                          vb_precomp, vp);

        /* 5. Process Odd Lanes: */
        __m256i c_odd = _shoup_avx_mullo(va_odd, vb,
                                         vb_precomp, vp);

        /* 6. Shift odd by 32 bits to the left results back
```

```
        and merge: */
        // [7, null, 5, null, 3, null, 1, null]
        __m256i res1_odd_shifted = _mm256_slli_epi64(c_odd, 32);

        // [7, null, 5, null, 3, null, 1, null]
        //or null, 6, null, 4, null, 2, null, 0]
        __m256i res_combined = _mm256_or_si256(c_even_,
                                                res_odd_shifted);

        /* 7. Store: */
        _mm256_storeu_si256((__m256i *)(res.elements + i),
                                        res_combined);
    }
    for (; i < size; i++)
        res.elements[i] = shoup(v.elements[i], param.b,
                                param.b_precomp, param.p);
    return res;
}
```

3.8.2 NEON

Listing 5: Auxilary Function of Shoup's Modular Reduction (Version Neon)

```
    TODO
```

Listing 6: NEON Implementation of Shoup's Modular Reduction

```
    TODO
```

3.9 Performance

3.9.1 Equipement detail:

The performance measurements and benchmarks were performed with the following hardware:

- Processor: AMD Ryzen™ 5 8600G Desktop Processor.
- Architecture: Zen 4 (supporting AVX, AVX2, and AVX-512 instruction sets)
- Compiler : GCC 13.3.0 with optimization flag -O3
- Execution : ./pcca7 -b_bits 31 (Option to set b to 31 bits)

3.9.2 Choice of unrolling

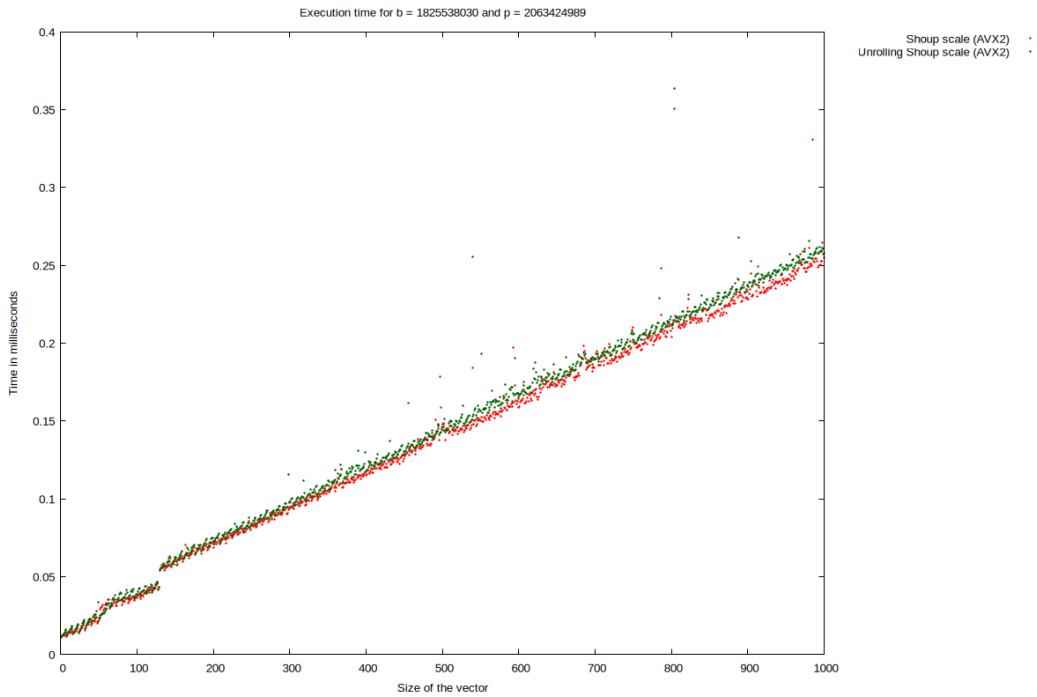

Figure 1: Execution Time Comparison: Standard vs. Unrolled

Table 1: Execution time of implementations with and without unrolling in milliseconds

Size	Standard	Unrolled
100	0.037	0.039
200	0.069	0.071
300	0.093	0.098
400	0.115	0.118
500	0.148	0.146
600	0.159	0.166
700	0.195	0.190
800	0.204	0.212
900	0.228	0.237
1000	0.255	0.258

Table 2: Average percentage speedup of the unrolling implementation relative to with unrolling

Size	Standard	Unrolled	Percentage difference
100	100%	105.41%	+5.41%
200	100%	102.90%	+2.90%
300	100%	105.38%	+5.38%
400	100%	102.61%	+2.61%
500	100%	98.65%	−1.35%
600	100%	104.40%	+4.40%
700	100%	97.44%	−2.56%
800	100%	103.92%	+3.92%
900	100%	103.95%	+3.95%
1000	100%	101.18%	+1.18%

Observation: On average, the Unrolled implementation is 2.58% slower than the standard version.

3.9.3 Choice between mullo_epi32 and mul_epu32

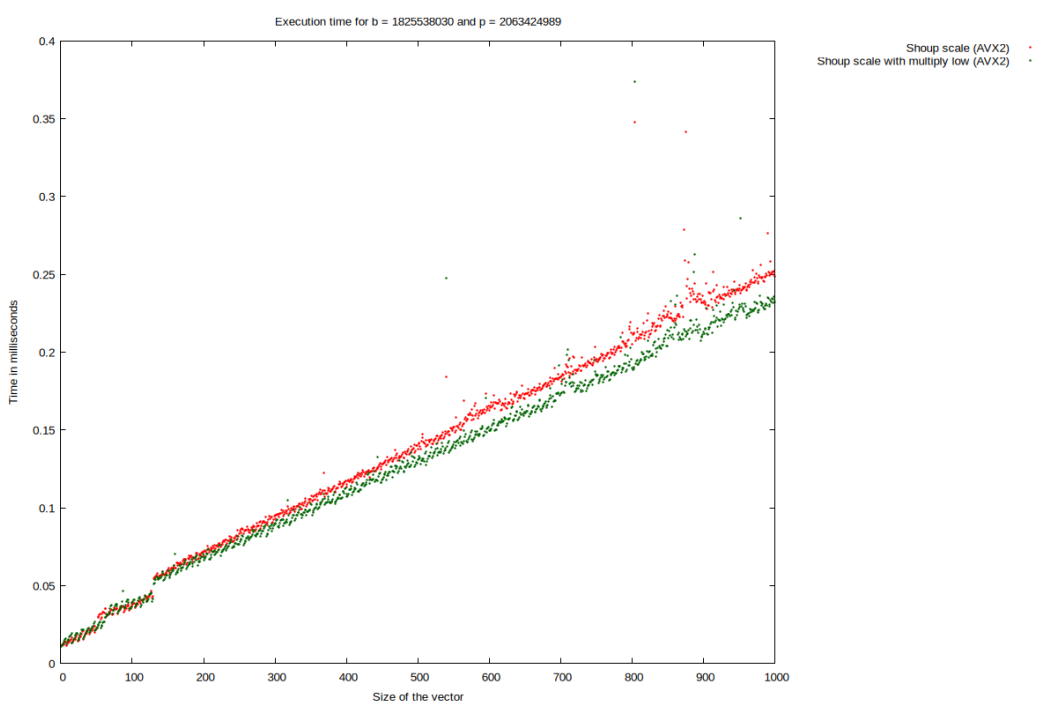

Figure 2: Execution Time Comparison: mullo_epi32 vs. mul_epu32

Table 3: Execution time of mullo_epi32 and mul_epu32 implementation in milliseconds

Size	mul_epu32	mullo_epi32
100	0.037	0.038
200	0.069	0.066
300	0.093	0.090
400	0.113	0.107
500	0.138	0.131
600	0.163	0.150
700	0.184	0.173
800	0.205	0.190
900	0.232	0.213
1000	0.249	0.232

Table 4: Average percentage speedup of mullo_epi32 implementation relative to mul_epu32

Size	mul_epu32	mullo_epi32	Percentage difference
100	100%	102.70%	+2.70%
200	100%	95.65%	−4.35%
300	100%	96.77%	−3.23%
400	100%	94.69%	−5.31%
500	100%	94.93%	−5.07%
600	100%	92.02%	−7.98%
700	100%	94.02%	−5.98%
800	100%	92.68%	−7.32%
900	100%	91.81%	−8.19%
1000	100%	93.17%	−6.83%

Observation: We can observe that at the smaller sizes (100), mullo_epi32 is slower than the standard mul_epu32 implementation by around 2%. But the effecientcy gain starts that show once the vector size hits 200 and above. Across all the vector sizes, the average percentage gain is 5.15%

3.9.4 Overall Performance

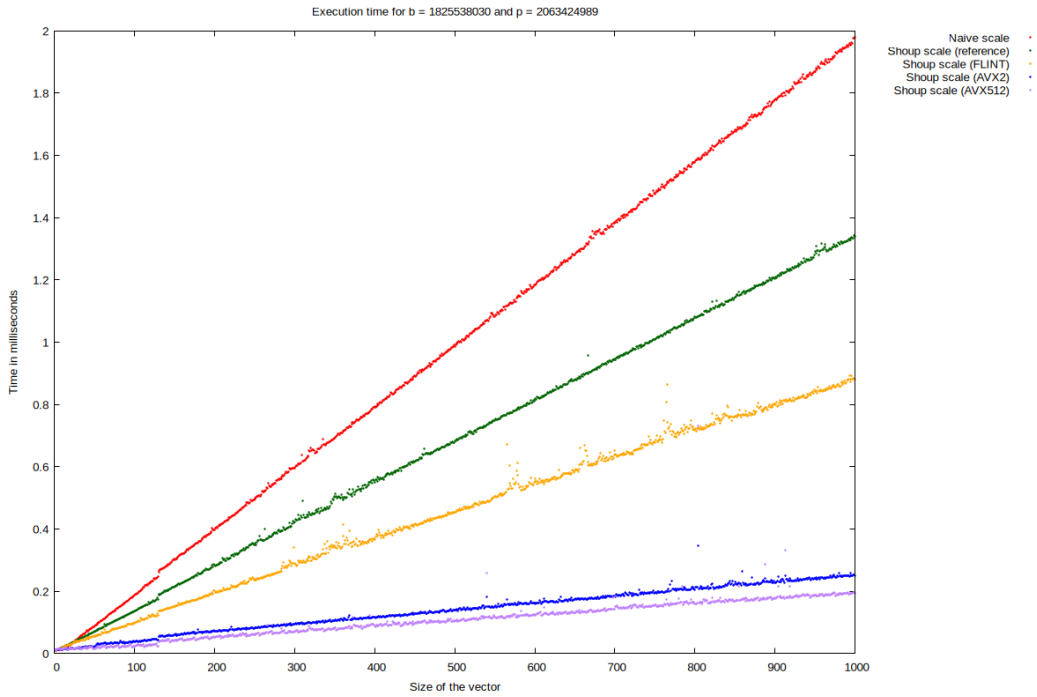

Figure 3: Execution Time Comparison Between Different Implementation For Modular Reduction

Table 5: Execution time for different implementations in relation too the size of the vector in milliseconds

Size	Naive	Shoup ref	FLINT	Shoup AVX2	Shoup AVX512
100	0.187	0.137	0.098	0.037	0.023
200	0.398	0.283	0.201	0.070	0.052
300	0.596	0.422	0.290	0.094	0.072
400	0.794	0.549	0.366	0.116	0.087
500	0.986	0.684	0.455	0.138	0.105
600	1.183	0.816	0.551	0.159	0.123
700	1.383	0.945	0.637	0.184	0.183
800	1.578	1.081	0.718	0.204	0.159
900	1.778	1.204	0.801	0.228	0.179
1000	1.979	1.339	0.880	0.249	0.193

Table 6: Average percentage speedup of different algorithms compared to the naive solution

Size	Naive	Shoup ref	FLINT	Shoup AVX2	Shoup AVX512
100	100%	73.26%	52.41%	19.79%	12.30%
200	100%	71.11%	50.50%	17.59%	13.07%
300	100%	70.81%	48.66%	15.77%	12.08%
400	100%	69.14%	46.10%	14.61%	10.96%
500	100%	69.37%	46.15%	14.00%	10.65%
600	100%	68.98%	46.58%	13.44%	10.40%
700	100%	68.33%	46.06%	13.30%	13.23%
800	100%	68.50%	45.50%	12.93%	10.08%
900	100%	67.72%	45.05%	12.82%	10.07%
1000	100%	67.66%	44.47%	12.58%	9.75%

Table 7: Relative Performance Summary in Percentages

Algorithm	Average %	1x
Naive	100.0%	Baseline
Shoup ref	68.7%	1.45x
FLINT	45.6%	2.2x
Shoup AVX2	15.4%	6.5x
Shoup AVX512	11.4%	8.7x

Observation: From the results retrieved, we can observe that FLINT's implementation over 64 bits provides roughly 2x faster modular reduction on 32 bit operations. Furthermore, our Shoup AVX2 and Shoup AVX512 implementations taillored to 32 bits integers provides significant faster 6-8x speedup.

3.9.5 NEON's Performance

TODO